Database Creation and Table Implementation

MySQL 8.0 DDL for the finalized nine-table hotel database

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Project: Hotel Management System. Output convention: result tables are expected outputs from the fixed rows in the shared SQL script; they are not screenshots of MySQL execution.

1. Database creation

This assignment creates the finalized MySQL schema. The same nine-table DDL is used by the shared SQL script and later reports, so foreign keys and query examples refer to one consistent design.

```
CREATE DATABASE IF NOT EXISTS hotel_management_assignment
  CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci;
USE hotel_management_assignment;
```

For a clean rebuild, the unified script drops and recreates only the assignment tables in reverse dependency order. Back up any data before rerunning it.

2. Table implementation

All tables use InnoDB. Integer primary keys identify entities; business identifiers such as guest number, email, service name, and room number receive unique constraints. Transaction and junction tables use foreign keys.

Table	Purpose	Main relationship
hotels	Hotel name, city, contact number	One hotel has many rooms and staff.
room_types	Room category, occupancy, base rate	One type describes many rooms.
guests	Guest identity and contact details	One guest can make many bookings.
staff	Employee details and salary	Each staff member belongs to one hotel.
rooms	Physical room and current status	Each room belongs to one hotel and one room type.
bookings	Guest stay dates and status	Each booking refers to one guest and one room.
services	Available paid services	One service can appear in many bookings.
booking_services	Services selected for a booking	Composite key prevents duplicate service links.
payments	Payment transaction and status	Each payment refers to one booking.

Core DDL example

```
CREATE TABLE bookings (
  booking_id INT UNSIGNED NOT NULL,
  guest_id INT UNSIGNED NOT NULL,
  room_id INT UNSIGNED NOT NULL,
  check_in DATE NOT NULL,
  check_out DATE NOT NULL,
  booking_status VARCHAR(16) NOT NULL DEFAULT 'CONFIRMED',
  PRIMARY KEY (booking_id),
  CONSTRAINT chk_booking_dates CHECK (check_out > check_in),
  CONSTRAINT chk_booking_status CHECK
    (booking_status IN ('CONFIRMED','CHECKED_IN','CHECKED_OUT','CANCELLED')),
  CONSTRAINT fk_bookings_guest FOREIGN KEY (guest_id)
    REFERENCES guests(guest_id) ON UPDATE CASCADE ON DELETE RESTRICT,
  CONSTRAINT fk_bookings_room FOREIGN KEY (room_id)
    REFERENCES rooms(room_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;
```

3. Expected schema check

```
USE hotel_management_assignment;
```

```
SHOW TABLES;
```

Expected table names
booking_services
bookings
guests
hotels
payments
room_types
rooms
services
staff

Run the unified SQL file in MySQL Workbench, then capture the schema name, the successful execution status, and SHOW TABLES output. MySQL 8.0.16 or later is recommended for enforced CHECK constraints.